

Set : $A = \{1, 2\}$; $B = \{1, 2, 2\}$ & $C = \{1, 2, 3\}$
 A & B are same sets
 $\downarrow$
 $(S, \oplus)$

Binary Operators :- If S is a non-empty set and $\otimes$ is defined on S , then $\otimes$ is said to be a Binary Operator if :-

$\otimes : S \times S \rightarrow S$: This Means Cross Maps S cross S to S
 Example :- $S = \mathbb{R}$ & $\otimes = +$, then $+$ is a Binary Operator.

Properties :-

1.) Commutative :- $a \otimes b = b \otimes a \quad \forall a, b \in S$
 Eg :- $S = \mathbb{R}$; $\otimes = +$ ✓✓
 $S = \mathbb{R}$; $\otimes = \times$ ✓✓
 But $S =$ set of Matrices & $\otimes = \times$ Not Commutative

2.) Associative :- $a \otimes (b \otimes c) = (a \otimes b) \otimes c$

3.) Distributivity :- $a \otimes (b \oplus c) = (a \otimes b) \oplus (a \otimes c)$
 $(a \oplus b) \otimes c = (a \otimes c) \oplus (b \otimes c) \rightarrow$ This is Left Distributivity
 This is Right Distributivity

Identity Element :- An element $e \in S$ is said to be left identity element if :-
 $e \otimes a = a \quad \forall a \in S$

Inverse of an Element :- b will be Inverse of a if :-
 $b \otimes a = e$ [Left Inverse]
 $a \otimes b = e$ [Right Inverse]

Group :- A group is an algebraic structure $(S, \otimes)$ which satisfies the following :-
 ① $a \otimes (b \otimes c) = (a \otimes b) \otimes c \quad \forall a, b, c \in S$
 ② $\forall a \in S \exists e$ such that $a \otimes e = e \otimes a = a$

③ $\forall a \in G \exists b \text{ s.t. } a \otimes b = b \otimes a = e$

Question :- If G is a group such that $(ab)^2 = a^2b^2 \forall a, b \in G$, then prove that G is an Abelian group $\rightarrow$ should be commutative

$$\begin{aligned} (ab)^2 &= a^2b^2 \Rightarrow (a \otimes b) \otimes (a \otimes b) = (a \otimes a) \otimes (b \otimes b) \\ \text{Let } c &= a \otimes b \in G; \text{ Then L.H.S.} \Rightarrow c \otimes (a \otimes b) \\ &= (c \otimes a) \otimes b = ((a \otimes b) \otimes a) \otimes b = (a \otimes (b \otimes a)) \otimes b \\ \text{Assuming } b \otimes a &= a \otimes b \rightarrow (a \otimes (a \otimes b)) \otimes b \\ &= ((a \otimes a) \otimes b) \otimes b = (d \otimes b) \otimes b = d \otimes (b \otimes b) \\ &= (a \otimes a) \otimes (b \otimes b) : \text{RHS} \end{aligned}$$

Hence our Assumption was correct $\Rightarrow G$ is an Abelian group

Groups, Rings & Fields

Qp. Verify whether a set of $n \times n$ non-singular matrices with real elements forms a group under matrix addition.

Let $A_{n \times n}$ be a non singular matrix with real elements. $(G, +)$

$$G = \left\{ \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} : a_{ij} \in \mathbb{R}, n \in \mathbb{N} \right\} \quad \det(G) \neq 0$$

① Test of Associativity - Consider $A, B, C \in G$

$$\begin{aligned} A + (B + C) &= \begin{bmatrix} a_{11} + (b_{11} + c_{11}) & \dots & a_{1n} + (b_{1n} + c_{1n}) \\ a_{21} + (b_{21} + c_{21}) & \dots & a_{2n} + (b_{2n} + c_{2n}) \\ \vdots & \ddots & \vdots \\ a_{n1} + (b_{n1} + c_{n1}) & \dots & a_{nn} + (b_{nn} + c_{nn}) \end{bmatrix} \\ &= (A + B) + C \quad \checkmark \end{aligned}$$

② Test of identity :- Consider $A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & \dots & \dots & a_{2n} \\ \vdots & \ddots & \ddots & \vdots \\ a_{n1} & \dots & \dots & a_{nn} \end{bmatrix} \in G$

Let $E = \begin{bmatrix} e_{11} & \dots & e_{1n} \\ e_{21} & \dots & e_{2n} \\ \vdots & \ddots & \vdots \\ e_{n1} & \dots & e_{nn} \end{bmatrix} \in G$ if possible; $[a_{ij}, e_{ij} \in \mathbb{R}]$

If $A + E = A$, then $\Rightarrow \begin{bmatrix} a_{11} + e_{11} & \dots & a_{1n} + e_{1n} \\ a_{21} + e_{21} & \dots & a_{2n} + e_{2n} \\ \vdots & \ddots & \vdots \\ a_{n1} + e_{n1} & \dots & a_{nn} + e_{nn} \end{bmatrix} = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ a_{21} & \dots & a_{2n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{nn} \end{bmatrix}$

For 2 matrices to be equal, their corresponding elements must be equal $\Rightarrow$ For ij^{th} element:-

$$\therefore \underline{E} = \begin{bmatrix} 0 & \dots & 0 \\ 0 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 0 \end{bmatrix}_{n \times n} \quad \begin{aligned} &\Rightarrow e_{ij} = 0 \text{ for } i, j \in n \\ &\text{(Null Matrix)} \end{aligned} \quad \text{But } \det(E) = 0 \Rightarrow E \notin G$$

$(G, +)$ is not a group.

This contradicts our assumption
So an identity element does not exist. $\Rightarrow$

Modification :- Let $\mathcal{L}$ now be the set of all $n \times n$ matrices, whether singular or not. Then, identity element, E exists & $E \in \mathcal{L}$.

② Test for Inverse : Let $\underline{A} = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix}$, $\underline{B} = \begin{bmatrix} b_{11} & \dots & b_{1n} \\ \vdots & \ddots & \vdots \\ b_{m1} & \dots & b_{mn} \end{bmatrix} \in \mathcal{L}$

such that $\underline{A} + \underline{B} = \underline{E}$, $\underline{E} = \begin{bmatrix} 0 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 0 \end{bmatrix} \in \mathcal{L}$ [Proved above]

$\therefore \begin{bmatrix} a_{11} + b_{11} & \dots & a_{1n} + b_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} + b_{m1} & \dots & a_{mn} + b_{mn} \end{bmatrix} = \begin{bmatrix} 0 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 0 \end{bmatrix}$; For ij^{th} element :- $a_{ij} + b_{ij} = 0$
 $\Rightarrow b_{ij} = -a_{ij} \quad \forall i, j \in \mathbb{R}$
 $\therefore \underline{B} = \begin{bmatrix} -a_{11} & \dots & -a_{1n} \\ \vdots & \ddots & \vdots \\ -a_{m1} & \dots & -a_{mn} \end{bmatrix} \rightarrow \underline{B} \in \mathcal{L}$ (as all its elements are real)
 $\therefore$ inverse exists.

Thus, $(\mathcal{L}, +)$ is a group when $\mathcal{L}$ = set of all $n \times n$ real matrices

Rings:

A ring is an algebraic structure $(R, +, *)$ which satisfies the following -

① $(R, +)$ is an Abelian group $\rightarrow$

a) $(a+b)+c = a+(b+c) \quad \forall a, b, c \in R$

b) $\forall a \in R \exists e \in R$ s.t. $a+e = e+a = a$

c) $\forall a \in R \exists b \in R$ s.t. $a+b = b+a = e$

d) $\forall a, b \in R, a+b = b+a$

② $(R, *)$ is a Monoid :- a) $(a*b)*c = a*(b*c) \quad \forall a, b, c \in R$
 Identity $\leftarrow$ b) $a*e' = e'*a = a \quad [\forall a \in R, \exists \text{ such } e' \in R]$

③ Distributivity of $*$ over $+$:- $(a+b)*c = (a*c) + (b*c)$
 $a*(b+c) = (a*b) + (a*c)$

Not included in this definition of a ring $\rightarrow$ a) Multiplicative Inverse
 b) Commutativity of Multiplication ; If ② is included we get a Field.
 But M.I. still unsure.

Note :- ① $a \otimes (b \otimes c) = (a \otimes b) \otimes c \rightarrow$ Semi group

$e \in \mathcal{L} \rightarrow$ Monoid $\xrightarrow{a^{-1}}$ group $\xrightarrow{a \otimes b = b \otimes a}$ Abelian group

② Sub-group :- $(\mathcal{L}, \otimes)$ $\mathcal{L} \otimes \mathcal{L} \rightarrow \mathcal{L}$
 $H \otimes H \rightarrow H$: This needed for a subgroup

Simply taking a subset of the group's set does not give a subgroup. Identity & Inverse may not be in H .

Qp. Verify whether a set of polynomials with real coefficients in a ring?
 $(R, +, *)$, $R = \left\{ \sum_{i=0}^n a_i x^i \mid a_i \in \mathbb{R}, n \in \mathbb{N} \cup \{0\} \right\}$

① Checking if $(R, +)$ is abelian $\rightarrow$ a) Let $A, B, C \in R$
 $A = a_0 + a_1 x + \dots + a_n x^n$; $B = b_0 + b_1 x + \dots + b_n x^n$] $a_i, b_i, c_i \in \mathbb{R}$
 $C = c_0 + c_1 x + \dots + c_n x^n$

Now, $A+B = (a_0+b_0) + (a_1+b_1)x + \dots + (a_n+b_n)x^n$
 $\Rightarrow (A+B)+C = ((a_0+b_0)+c_0) + ((a_1+b_1)+c_1)x + \dots + ((a_n+b_n)+c_n)x^n$
 Since Associativity for real numbers is known, $(a_i+b_i)+c_i = a_i+(b_i+c_i)$
 $\therefore (A+B)+C = ((a_0+b_0)+c_0) + \dots + ((a_n+b_n)+c_n)x^n$
 $= (a_0+(b_0+c_0)) + \dots + (a_n+(b_n+c_n))x^n = A+(B+C)$

Hence Associativity is proven for $(R, +)$
 Rest can be proven

Linear Vector Spaces

Let V be a set of ordered pairs (a, b) , $a, b \in \mathbb{R}$. The following operations are defined -

$(a, b) \otimes (c, d) = (a+c, b+d)$, $k \odot (a, b) = (ka, 0)$
 I identify whether V is a vector space over $\mathbb{R}$

An algebraic structure $(V, F, \otimes, \odot)$ equipped with a Non Empty set V , Field F , a binary operation $\otimes: V \times V \rightarrow V$ and an External mapping $\odot: F \times V \rightarrow V$ is called a space if:-

① $(V, \otimes)$ is an Abelian group

② $a \odot (u \otimes v) = (a \odot u) \otimes (a \odot v)$
 $(u \otimes v) \odot b = (u \odot b) \otimes (v \odot b)$

Here, $u, v \in V$ & $a, b \in F$

Qp. If $V = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, x_i \in \mathbb{R}, n \in \mathbb{I}^+ \right\}$ $\otimes = +$ & $\odot = *$, $F = \mathbb{R}$
 Eq. of Vector space

① Let $a = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$, $b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$ & $c = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$, $a, b, c \in V$

Now, $(a+b) = \begin{bmatrix} a_1+b_1 \\ a_2+b_2 \\ \vdots \\ a_n+b_n \end{bmatrix} \Rightarrow (a+b)+c = \begin{bmatrix} (a_1+b_1)+c_1 \\ (a_2+b_2)+c_2 \\ \vdots \\ (a_n+b_n)+c_n \end{bmatrix} = \begin{bmatrix} a_1+(b_1+c_1) \\ a_2+(b_2+c_2) \\ \vdots \\ a_n+(b_n+c_n) \end{bmatrix}$
 $= a+(b+c) \rightarrow$ A associativity

Rest can be proven $\Rightarrow$ Abelian group

② Let $x, y \in \mathbb{R}$ & $A, B \in V \rightarrow$

Now, $x * (A+B) = x * \begin{bmatrix} a_1+b_1 \\ a_2+b_2 \\ \vdots \\ a_n+b_n \end{bmatrix} = \begin{bmatrix} x a_1 + x b_1 \\ x a_2 + x b_2 \\ \vdots \\ x a_n + x b_n \end{bmatrix}$

$$= \begin{bmatrix} x a_1 \\ x a_2 \\ \vdots \end{bmatrix} + \begin{bmatrix} x b_1 \\ x b_2 \\ \vdots \end{bmatrix}$$

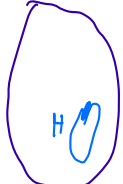
$$= (x * A) + (x * B)$$

Root can also be shown

★ A vector space will be **Linear** based on the operators $\oplus$ & $\odot$

★★ $\Rightarrow$ Let $u = (a, b)$, $v = (c, d)$, $w = (e, f)$, where $u, v, w \in V$
 Now, $u \oplus v = (a, b) \oplus (c, d) = (a+c, b+d)$
 $\Rightarrow (u \oplus v) \oplus w = (a+c, b+d) \oplus (e, f) = (a+c+e, b+d+f)$
 $= (a + (c+e), b + (d+f)) = u \oplus (v \oplus w) \rightarrow$ **Associativity**
 Root can be shown $\Rightarrow (V, \oplus)$ is an Abelian group

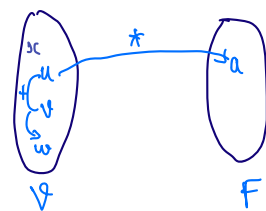
③ Let $u = (a, b)$, $v = (c, d)$; $u, v \in V$
 Let $k \in \mathbb{R}$
 Now, $k \odot (u \oplus v) = k \odot (a+c, b+d) = (k(a+c), k(b+d)) = (ka+kc, kb+kd)$
 Now, $k \odot u \oplus k \odot v = (ka, kb) \oplus (kc, kd) = (ka+kc, kb+kd)$
 Right Distributivity also True $\Rightarrow V$ is indeed a vector space over $\mathbb{R}$

Sub Space :- V  Q. Let S be set of all elements of the form :-
 $(x+2y, y, -x+3y) \in \mathbb{R}^3$ where $x, y \in \mathbb{R}$
 Show that S is sub space of $\mathbb{R}^3$

1.) $u + v \in S$ 2.) $(\alpha u) \in S$
 Let $u = (x_1 + 2y_1, y_1, -x_1 + 3y_1)$ & $v = (x_2 + 2y_2, y_2, -x_2 + 3y_2)$
 $\therefore u + v = (x_1 + 2y_1 + x_2 + 2y_2, y_1 + y_2, -x_1 + 3y_1 - x_2 + 3y_2)$
 $= (x + 2y, y, -x + 3y) \Rightarrow$ Belongs to S

Linear Combination Operation

$$\underline{w} = a \underline{u} + b \underline{v} \quad [\text{Linear Combination}]$$



$$+ : V \times V \rightarrow V$$

$$* : F \times V \rightarrow V$$

a) Express the polynomial $p(x)$ as a L.C. of ϕ_i 's.
 $\phi_2(x) = 2x^2 - 3x$ & $\phi_3(x) = x + 3$

$$p(x) = x^2 + 4x - 3; \quad \phi_1(x) = x^2 - 2x + 5$$

Let $p = a \phi_1 + b \phi_2 + c \phi_3$ where $a, b, c \in F \Rightarrow x^2 + 4x - 3 = a(x^2 - 2x + 5) + b(2x^2 - 3x) + c(x + 3)$
 Comparing coefficients of x^2, x & $1 \rightarrow$
 $1 = a + 2b$ $4 = -2a - 3b + c$ & $-3 = 5a + 3c$
 Solving $\rightarrow \begin{bmatrix} 1 & 2 & 0 & | & 1 \\ -2 & -3 & 1 & | & 4 \\ 5 & 0 & 3 & | & -3 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ -3 \end{bmatrix} \equiv \begin{bmatrix} 1 & 2 & 0 & | & 1 \\ -2 & -3 & 1 & | & 4 \\ 5 & 0 & 3 & | & -3 \end{bmatrix}$

$$\begin{bmatrix} - & -3 & 1 & | & 4 \end{bmatrix} R_2 = R_2 - 3R_1 \quad \begin{bmatrix} -2 & -3 & 1 & | & 4 \end{bmatrix} R_2 = R_2 - 11R_3 \quad \begin{bmatrix} -2 & -3 & 1 & | & 4 \end{bmatrix}$$

$$\begin{aligned} \Leftrightarrow \left[\begin{array}{ccc|c} 2 & 5 & 3 & -3 \\ 1 & 2 & 0 & 1 \end{array} \right] & \Leftrightarrow \left[\begin{array}{ccc|c} 11 & 9 & 0 & -15 \\ 1 & 2 & 0 & 1 \end{array} \right] & \Leftrightarrow \left[\begin{array}{ccc|c} 0 & -13 & 0 & -26 \\ 12 & 0 & 0 & 1 \end{array} \right] \end{aligned}$$

$$\therefore b = 2, \quad a = -3 \quad \& \quad c = 4$$

b.) Consider $\underline{v}_1 = [1 \ 2 \ 3]^T$, $\underline{v}_2 = [2 \ 3 \ 1]^T$ & $\underline{v} = [a \ b \ c]^T$. Find the conditions on a, b & c so that $\underline{v}$ is a L.C. of $\underline{v}_1$ & $\underline{v}_2$.

$$\begin{aligned} \text{Let } \underline{v} &= p \underline{v}_1 + q \underline{v}_2 \quad ; \quad p, q \in F \quad \Rightarrow \quad \begin{pmatrix} a \\ b \\ c \end{pmatrix} = p \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + q \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} \\ \Rightarrow \left[\begin{array}{cc|c} 1 & 2 & a \\ 2 & 3 & b \\ 3 & 1 & c \end{array} \right] \begin{bmatrix} p \\ q \end{bmatrix} &= \begin{bmatrix} a \\ b \\ c \end{bmatrix} \quad \Rightarrow \quad \left[\begin{array}{cc|c} 1 & 2 & a \\ 2 & 3 & b \\ 3 & 1 & c \end{array} \right] \begin{bmatrix} p \\ q \end{bmatrix} \xrightarrow{R_3 = R_3 - 3R_1} \left[\begin{array}{cc|c} 1 & 2 & a \\ 2 & 3 & b \\ 0 & -5 & c-3a \end{array} \right] \\ \xrightarrow{R_2 = R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 2 & a \\ 0 & -1 & b-2a \\ 0 & -5 & c-3a \end{array} \right] & \xrightarrow{R_3 = R_3 - 5R_2} \left[\begin{array}{cc|c} 1 & 2 & a \\ 0 & -1 & b-2a \\ 0 & 0 & 7a-5b+c \end{array} \right] \end{aligned}$$

$$\text{Thus, } 7a - 5b + c = 0$$

c.) Show that the vectors $\underline{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ & $\underline{v}_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ span $\mathbb{R}^2$ over $\mathbb{R}$

Let $a, b \in \mathbb{R}$; Let $\underline{v} = a \underline{v}_1 + b \underline{v}_2$ [Linear Combination of vectors in question]

$$\therefore \underline{v} = \begin{pmatrix} a \\ a \end{pmatrix} + \begin{pmatrix} b \\ 2b \end{pmatrix} = \begin{pmatrix} a+b \\ a+2b \end{pmatrix} = \begin{pmatrix} p \\ q \end{pmatrix} \quad (\text{Assume}) ; \quad p, q \in \mathbb{R}$$

$$\begin{aligned} \text{Now, } \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} &= \begin{pmatrix} p \\ q \end{pmatrix} \quad \Rightarrow \quad \left[\begin{array}{cc|c} 1 & 1 & p \\ 1 & 2 & q \end{array} \right] \xrightarrow{R_2 = R_2 - R_1} \left[\begin{array}{cc|c} 1 & 1 & p \\ 0 & 1 & q-p \end{array} \right] \\ \xrightarrow{R_1 = R_1 - R_2} \left[\begin{array}{cc|c} 1 & 0 & 2p-q \\ 0 & 1 & q-p \end{array} \right] & , \quad \text{so } a = 2p-q \quad \& \quad b = q-p \end{aligned}$$

Thus, $\exists a, b \in \mathbb{R} \quad \forall p, q \in \mathbb{R} \Rightarrow \underline{v}_1, \underline{v}_2$ span $\mathbb{R}^2$ [i.e. any $\underline{v} \in \mathbb{R}^2$ over $\mathbb{R}$]

$$[2p-q \in \mathbb{R} \quad \& \quad q-p \in \mathbb{R} \quad \because p, q \in \mathbb{R}]$$

Linear Independence, Basis, Dimensions

$$\text{Qp. } f(x) = 2x^3 + x^2 + x + 1, \quad g(x) = x^3 + 3x^2 + x - 2 \quad \& \quad h(x) = x^3 + 2x^2 - x + 3$$

$$a f(x) + b g(x) + c h(x) = 0 \Rightarrow x^3 [2a + b + c] + x^2 [a + 3b + 2c] + x [a + b - c] + 1 [a - 2b + 3c] = 0$$

Thus $\rightarrow \left[\begin{array}{ccc|c} 2 & 1 & 1 & 0 \\ 1 & 3 & 2 & 0 \\ 1 & 1 & -1 & 0 \\ 1 & -2 & 3 & 0 \end{array} \right]$, On solving: $\left[\begin{array}{ccc|c} 1 & 0.5 & 0.5 & 0 \\ 0 & 6 & 0 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 2 & 0 \end{array} \right]$

So, $2c = 0 \Rightarrow c = 0$ & $6b = 0 \Rightarrow b = 0$; $a = 0$

$\therefore a = b = c = 0 \Rightarrow$ Linear Independence

Qp. Check linear independence of $f(x)$ & $g(x)$ Let $a f(x) + b g(x) = 0$

Let $a f(x) + b g(x) = 0 \Rightarrow a x + b e^{2x} = 0$

Wronskian Determinant [If it is ^{not 0}, then L.I., otherwise not]

$\begin{vmatrix} f & g \\ f' & g' \end{vmatrix} = \begin{vmatrix} x & e^{2x} \\ 1 & 2e^{2x} \end{vmatrix} = 2x e^{2x} - e^{2x} = e^{2x} [2x - 1] \neq 0$ in general

So, Linearly Independent

Now, $h(x) = \sin(x) \Rightarrow W = \begin{vmatrix} f & g & h \\ f' & g' & h' \end{vmatrix} = \begin{vmatrix} x & e^{2x} & \sin x \\ 1 & 2e^{2x} & \cos x \\ 0 & 2e^{2x} & -\sin x \end{vmatrix}$

$= x [-2\sin(x)e^{2x} - 4\cos(x)e^{2x}] - e^{2x} [-\sin(x)] + \sin(x) [4e^{2x}]$

$= -2x \sin(x)e^{2x} - 4x \cos(x)e^{2x} + 5\sin(x)e^{2x} = e^{2x} [5\sin(x) - 2x(\sin(x) + 2\cos(x))] \neq 0$ in general

So Linearly Independent

Basis & Dimensions :-

$(3, 1, 2) = 3(1, 0, 0) + 1(0, 1, 0) + 2(0, 0, 1)$

Similarly, $(x_i, y_i, z_i) = x_i(1, 0, 0) + y_i(0, 1, 0) + z_i(0, 0, 1)$

Any random vector (x_i, y_i, z_i) can be written as a suitable L.C. of $(1, 0, 0), (0, 1, 0), (0, 0, 1)$ form. All these are Linearly independent also. Thus, $(1, 0, 0), (0, 1, 0)$ & $(0, 0, 1)$ form a Basis of $\mathbb{R}^3$ over $\mathbb{R}$.

Basis is a Linearly Independent set of vectors which span the whole vector space

Cardinality of Basis = Dimension of Vector space

Qp. Determine whether $\underline{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$, $\underline{v}_2 = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 2 \end{bmatrix}$, $\underline{v}_3 =$

$\begin{bmatrix} 2 \\ 5 \\ 6 \\ 4 \end{bmatrix}$ & $\underline{v}_4 = \begin{bmatrix} 2 \\ 6 \\ 8 \\ 5 \end{bmatrix}$ form a basis of $\mathbb{R}^4$ over $\mathbb{R}$. If not, determine the dimension of the subspace they span.

Let $\underline{v} = a \underline{v}_1 + b \underline{v}_2 + c \underline{v}_3 + d \underline{v}_4$

$[a + b + 2c + 2d] \quad [e]$

$$\therefore V = \begin{bmatrix} a+2b+5c+6d \\ a+3b+6c+8d \\ a+2b+7c+5d \end{bmatrix} = \begin{bmatrix} p \\ q \\ h \\ b \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 2 & 2 \\ 1 & 2 & 5 & 6 \\ 1 & 3 & 6 & 8 \\ 1 & 2 & 4 & 5 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} p \\ q \\ h \\ b \end{bmatrix} \equiv \begin{bmatrix} 1 & 1 & 2 & 2 & | & p \\ 1 & 2 & 5 & 6 & | & q \\ 1 & 3 & 6 & 8 & | & h \\ 1 & 2 & 4 & 5 & | & b \end{bmatrix}$$

$$\leftrightarrow \begin{bmatrix} 1 & 1 & 2 & 2 & | & p \\ 0 & 1 & 3 & 4 & | & q-p \\ 0 & 2 & 4 & 6 & | & h-p \\ 0 & 1 & 2 & 3 & | & b-p \end{bmatrix} \leftrightarrow \begin{bmatrix} 1 & 1 & 2 & 2 & | & p \\ 0 & 1 & 3 & 4 & | & q-p \\ 0 & 0 & -2 & -2 & | & h+p-2q \\ 0 & 0 & -1 & -1 & | & b-q \end{bmatrix}$$

$$\leftrightarrow \begin{bmatrix} 1 & 1 & 2 & 2 & | & p \\ 0 & 1 & 3 & 4 & | & q-p \\ 0 & 0 & -1 & -1 & | & b-q \\ 0 & 0 & 0 & 0 & | & h+p-2b \end{bmatrix}$$

Not Linearly Independent
Dimension = 3

Null & Range Space

Q1) $x+y=2$; $x+2y+z=4$; $y+2z+t=4$; $z+2t=3$

Q2) $x_1+x_2+x_3=1$; $2x_1+3x_2+x_3=2$; $5x_1+6x_2+4x_3=1$

$$1.) \begin{bmatrix} 1 & 1 & 0 & 0 & | & 2 \\ 1 & 2 & 1 & 0 & | & 4 \\ 0 & 1 & 2 & 1 & | & 4 \\ 0 & 0 & 1 & 2 & | & 3 \end{bmatrix} \leftrightarrow \begin{bmatrix} 1 & 1 & 0 & 0 & | & 2 \\ 0 & 1 & 1 & 0 & | & 2 \\ 0 & 1 & 2 & 1 & | & 4 \\ 0 & 0 & 1 & 2 & | & 3 \end{bmatrix} \leftrightarrow \begin{bmatrix} 1 & 1 & 0 & 0 & | & 2 \\ 0 & 1 & 1 & 0 & | & 2 \\ 0 & 0 & 1 & 1 & | & 2 \\ 0 & 0 & 1 & 2 & | & 3 \end{bmatrix} \leftrightarrow \begin{bmatrix} 1 & 1 & 0 & 0 & | & 2 \\ 0 & 1 & 1 & 0 & | & 2 \\ 0 & 0 & 1 & 1 & | & 2 \\ 0 & 0 & 0 & 1 & | & 1 \end{bmatrix}$$

$\therefore x=y=z=t=1$ (1)

$$2.) \begin{bmatrix} 1 & 1 & 1 & | & 1 \\ 2 & 3 & 1 & | & 2 \\ 5 & 6 & 1 & | & 1 \end{bmatrix} \leftrightarrow \begin{bmatrix} 1 & 1 & 1 & | & 1 \\ 0 & 1 & -1 & | & 0 \\ 0 & 1 & -1 & | & -4 \end{bmatrix}$$

No solution (2)

① If there exists only a trivial solution ($x=0$) to equation $Ax=0$, then a unique solution will exist for $Ax=b$

Eg :- $\begin{bmatrix} 1 & 1 & 0 & 0 & | & 0 \\ 1 & 2 & 1 & 0 & | & 0 \\ 0 & 1 & 2 & 1 & | & 0 \\ 0 & 0 & 1 & 2 & | & 0 \end{bmatrix} \leftrightarrow \begin{bmatrix} 1 & 1 & 0 & 0 & | & 0 \\ 0 & 1 & 1 & 0 & | & 0 \\ 0 & 0 & 1 & 1 & | & 0 \\ 0 & 0 & 0 & 1 & | & 0 \end{bmatrix} \Rightarrow x=y=z=t=0$

In general $\rightarrow \begin{bmatrix} 1 & 1 & 0 & 0 & | & b_1 \\ 0 & 1 & 1 & 0 & | & b_2-b_1 \\ 0 & 0 & 1 & 1 & | & b_3-b_2+b_1 \\ 0 & 0 & 0 & 1 & | & b_4-b_3+b_2-b_1 \end{bmatrix}$

In case of ② $\rightarrow \left[\begin{array}{ccc|c|c|c} 1 & 1 & 1 & 0 & 1 & b_1 \\ 2 & 3 & 1 & 0 & 2 & b_2 \\ 5 & 6 & 4 & 0 & 1 & b_3 \end{array} \right] \leftrightarrow \left[\begin{array}{ccc|c|c|c} 1 & 1 & 1 & 0 & 1 & b_1 \\ 0 & 1 & -1 & 0 & 0 & b_2 - 2b_1 \\ 0 & 1 & -1 & 0 & -4 & b_3 - 5b_1 \end{array} \right]$

$\leftrightarrow \left[\begin{array}{ccc|c|c|c} 1 & 1 & 1 & 0 & 1 & b_1 \\ 0 & 1 & -1 & 0 & 0 & b_2 - b_1 \\ 0 & 0 & 0 & 0 & -4 & b_3 - b_2 - 3b_1 \end{array} \right]$ Let $x_3 = \alpha = x_2$
 $\therefore x_1 = -2\alpha \Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \alpha \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$

In general, ② will have solution only if $b_3 - b_2 - 3b_1 = 0$ Dimension = 1
[Null space]

Rank(A) = 2 & Rank(A|b) = 2 only if $\uparrow$

Set $b_1 = \beta$ & $b_2 = \gamma \Rightarrow b_3 = 3\beta + \gamma$; $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} \beta \\ \gamma \\ 3\beta + \gamma \end{bmatrix} = \beta \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} + \gamma \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$
 Dimension = 2 [Range space]

Q.3) $A = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 1 & 2 \\ 4 & -1 & 9 \\ 2 & 1 & 1 \end{bmatrix}$ & $B = [3 \ 0 \ 0 \ 2]^T$

$A|B = \left[\begin{array}{ccc|c|c|c} 1 & -1 & 2 & 0 & 3 & b_1 \\ 2 & 1 & 2 & 0 & 0 & b_2 \\ 4 & -1 & 9 & 0 & 0 & b_3 \\ 2 & 1 & 1 & 0 & 2 & b_4 \end{array} \right] \leftrightarrow \left[\begin{array}{ccc|c|c|c} 1 & -1 & 2 & 0 & 3 & b_1 \\ 0 & 3 & -2 & 0 & -6 & b_2 - 2b_1 \\ 0 & 3 & 1 & 0 & -12 & b_3 - 4b_1 \\ 0 & 3 & -3 & 0 & -4 & b_4 - 2b_1 \end{array} \right]$

$\leftrightarrow \left[\begin{array}{ccc|c|c|c} 1 & -1 & 2 & 0 & 3 & b_1 \\ 0 & 3 & -2 & 0 & -6 & b_2 - 2b_1 \\ 0 & 0 & 3 & 0 & -6 & b_3 - b_2 - 2b_1 \\ 0 & 0 & -1 & 0 & 2 & b_4 - b_2 \end{array} \right] \leftrightarrow \left[\begin{array}{ccc|c|c|c} 1 & -1 & 2 & 0 & 3 & b_1 \\ 0 & 3 & -2 & 0 & -6 & b_2 - 2b_1 \\ 0 & 0 & 3 & 0 & -6 & b_3 - b_2 - 2b_1 \\ 0 & 0 & 0 & 0 & 0 & 3b_4 + b_3 - 4b_2 - 2b_1 \end{array} \right]$

$Ax = 0 \Rightarrow [x_1 \ x_2 \ x_3]^T = 0 \rightarrow$ Null space

Condition for solvability $\rightarrow 3b_4 + b_3 - 4b_2 - 2b_1 = 0$
 Set $b_1 = \alpha$, $b_2 = \beta$ & $b_3 = \gamma \Rightarrow b_4 = \frac{2}{3}\alpha + \frac{1}{3}\beta - \frac{\gamma}{3}$

$\Rightarrow \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ 0 \\ 0 \\ 2/3 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1/3 \end{bmatrix} + \gamma \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1/3 \end{bmatrix}$

Range space
Dimension = 3

Linear Transformation

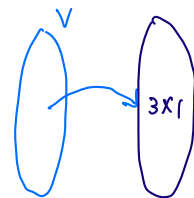
Why Transform?
To change space / Domain
 $f(x) = \sin(x)$; $\mathcal{L}\{f\} = \frac{1}{s^2+1}$
(x-space to s-space)

$$\underline{v}_1 = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} \quad \& \quad \underline{v}_2 = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} ; \quad a, b \in F$$

$$\underline{v}_3 = a \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} \quad , \quad \underline{v}_4 = a \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} + b \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} ; \quad \underline{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$\underline{A} \underline{v}_1 = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} L_1 \\ L_2 \end{bmatrix} \rightarrow \text{Still in same Vector space}$$

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix} \quad \text{Space has been Transformed}$$



$$\text{Transformation, } t : \mathbb{R}^{2 \times 1} \rightarrow \mathbb{R}^{3 \times 1}$$

When the image is in the same vector space, we call it Operator / operation. If image is in a new vector space, we call it Transformation.

$$\left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} ; x_i \in \mathbb{R} \right\} \rightarrow \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} ; x_i \in \mathbb{R} \right\}$$

Example :- $S = \{ax \mid a \in \mathbb{R}, a > 0\}$ & $V = \{a \mid a \in \mathbb{R}, a > 0\}$ New Set
d/dx $\therefore$ Transformation

★ Linearity Condition - ① $t(\underline{v}_1 + \underline{v}_2) = t(\underline{v}_1) + t(\underline{v}_2)$ ② $t(\alpha \underline{v}_1) = \alpha t(\underline{v}_1)$
Combining these two $\rightarrow t(\alpha \underline{v}_1 + \beta \underline{v}_2) = \alpha t(\underline{v}_1) + \beta t(\underline{v}_2)$

Qp-1 Let V be a Real Vector Space with the following operator defined - $T: V \rightarrow V$
 $T[f(x)] = \int_0^x f(t) dt \quad \forall f(x) \in V, x \in \mathbb{R}$ Check if T is a Linear Transformation or not.

Let $f_1(x)$ & $f_2(x) \in V$; $T[\alpha f_1(x) + \beta f_2(x)] = \int_0^x (\alpha f_1(t) + \beta f_2(t)) dt = \alpha \int_0^x f_1(t) dt + \beta \int_0^x f_2(t) dt$
 $= \alpha T[f_1(x)] + \beta T[f_2(x)] \therefore T$ is a Linear Transformation

Qp-2 I identify whether the following mappings are linear transformations.

a.) $t: \mathbb{R}^3 \rightarrow \mathbb{R}^2$; $t(x, y, z) = (x + 2y - 3z, 4x - 5y + 6z)$
b.) $t: \mathbb{R}^3 \rightarrow \mathbb{R}^2$; $t(x, y, z) = (x+1, y+1)$

a.) Let (x_1, y_1, z_1) & $(x_2, y_2, z_2) \in \mathbb{R}^3$; $t[(x_1, y_1, z_1) + (x_2, y_2, z_2)]$
 $= t(x_1 + x_2, y_1 + y_2, z_1 + z_2) = [x_1 + x_2 + 2y_1 + 2y_2, 4x_1 + 4x_2 - 5y_1 - 5y_2 + 6z_1 + 6z_2]$
 $= [x_1 + 2y_1 - 3z_1, 4x_1 - 5y_1 + 6z_1] + [x_2 + 2y_2 - 3z_2, 4x_2 - 5y_2 + 6z_2]$

$$\begin{aligned}
 &= t(x_1, y_1, z_1) + t(x_2, y_2, z_2) \quad ; \text{ And } \alpha \in \mathbb{R} \rightarrow t[\alpha(x_1, y_1, z_1)] \\
 &= t(\alpha x_1, \alpha y_1, \alpha z_1) = (\alpha x_1 + 2\alpha y_1 - 3\alpha z_1, 4\alpha x_1 - 5\alpha y_1 + 6\alpha z_1) \\
 &= [\alpha(x_1 + 2y_1 - 3z_1), \alpha(4x_1 - 5y_1 + 6z_1)] = \alpha t[x_1, y_1, z_1]
 \end{aligned}$$

Hence t is a Linear Transformation

$$\begin{aligned}
 \text{b) Let } (x_1, y_1, z_1), (x_2, y_2, z_2) &\in \mathbb{R}^3 \rightarrow t[(x_1, y_1, z_1) + (x_2, y_2, z_2)] \\
 &= t[(x_1 + x_2, y_1 + y_2, z_1 + z_2)] = (x_1 + x_2 + 1, y_1 + y_2 + 1) \\
 \text{Now } t(x_1, y_1, z_1) + t(x_2, y_2, z_2) &= (x_1 + 1, y_1 + 1) + (x_2 + 1, y_2 + 1) \\
 &= (x_1 + x_2 + 2, y_1 + y_2 + 2) \Rightarrow \text{t is Not Linear}
 \end{aligned}$$

Qp.3 Check linearity of these operators $\rightarrow$ a.) $\hat{L} = \frac{d^2}{dx^2} - m^2$, b.) $\hat{L} = \frac{\partial}{\partial t} - \alpha \frac{\partial^2}{\partial z^2} + \beta$

$$\begin{aligned}
 \text{a) Let } f(x) \text{ \& } g(x) &\in V \quad \text{Let } p, q \in \mathbb{R} \quad ; \text{ Now } \rightarrow \hat{L}(pf(x) + qg(x)) \\
 &= \frac{d^2}{dx^2} [pf(x) + qg(x)] - m^2 [pf(x) + qg(x)] = p \left[\frac{d^2}{dx^2} f - m^2 f \right] + q \left[\frac{d^2}{dx^2} g - m^2 g \right] \\
 &= p \hat{L}[f(x)] + q \hat{L}[g(x)] \quad \therefore \hat{L} \text{ is linear}
 \end{aligned}$$

$$\begin{aligned}
 \text{b) Let } f(t, z) \text{ \& } g(t, z) &\in V \quad \text{\& } a, b \in \mathbb{R} \quad ; \text{ Now } \rightarrow \hat{L}(af(t, z) + bg(t, z)) \\
 &= \frac{\partial}{\partial t} [af + bg] - \alpha \frac{\partial^2}{\partial z^2} [af + bg] + \beta [af + bg] = a \frac{\partial f}{\partial t} + b \frac{\partial g}{\partial t} - \alpha a \frac{\partial^2 f}{\partial z^2} - \alpha b \frac{\partial^2 g}{\partial z^2} + \beta a f + \beta b g \\
 &= a \left[\frac{\partial}{\partial t} - \alpha \frac{\partial^2}{\partial z^2} + \beta \right] f(t, z) + b \left[\frac{\partial}{\partial t} - \alpha \frac{\partial^2}{\partial z^2} + \beta \right] g(t, z) = a \hat{L}(f) + b \hat{L}(g) \\
 &\quad \text{Hence Linear}
 \end{aligned}$$

Qp.4) Let $B = \left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$ be a basis for $\mathbb{R}^3$ (IR)

$$t \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad t \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad \& \quad t \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} \quad \text{such that} \quad \text{Find the value of } t \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$$

$$\text{Let } \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix} = a \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} + c \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix} = \begin{bmatrix} c - a \\ b - c \\ a - b + c \end{bmatrix}$$

$$\therefore a = 2, \quad b = 0 \quad \& \quad a = 1 \quad ; \quad t \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix} = 1 \times t \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + 2 \times t \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 2 \\ -4 \\ 6 \end{bmatrix} = \begin{bmatrix} 3 \\ -4 \\ 6 \end{bmatrix}$$